Environmental Innovation and Societal Transitions 50 (2024) 100812

Contents lists available at ScienceDirect

Environmental Innovation and Societal Transitions

journal homepage: www.elsevier.com/locate/eist

Transition conflicts: A Gramscian political ecology perspective on the contested nature of sustainability transitions

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ARTICLE INFO

Keywords:
Sustainability transitions
Conflicts
Power relations
Political ecology
Gramsci

ABSTRACT

Despite a broad consensus on sustainability, conflicts are increasingly prevalent in sustainability transitions. Although these conflicts significantly influence transition dynamics and socio-ecological futures, the role of conflicts in sustainability transitions remains insufficiently addressed. This paper aims to elucidate the contested dynamics of sustainability transitions by merging political ecology's emphasis on conflicts, nature, power, and justice with Gramscian hegemony theory. The integrated framework of Gramscian political ecology enables the analysis of transition conflicts as struggles for hegemony on the terrain of society-nature relations amid ecological crises. A brief comparative study of coal transitions in South Africa and Germany serves to illustrate the key insights that Gramscian political ecology offers into the contested nature of sustainability transitions, including conflict dynamics, power strategies, and barriers and potentials for radical transformative change.

1. Introduction

Despite a broad consensus on sustainability and climate action, conflicts are increasingly prevalent in sustainability transitions. With escalating ecological crises and increasing transition pressures, conflicts – like those regarding fossil fuel phase-out (Blondeel, 2019), renewable energy infrastructure (Dunlap, 2018) and industrial transformation (Pichler et al., 2021) are expected to multiply and intensify. The trajectory of socio-ecological change is significantly influenced by conflicts among collective actors with competing interests, beliefs, and goals who operate within a context of uneven power relations. I refer to these as transition conflicts[1], defined as struggles for hegemony that emerge on the terrain of society-nature relations in response to the politicization of ecological crises. Recognizing that social struggles drive social change, this paper emphasizes the crucial role of transition conflicts as both potential catalysts and impediments to sustainability transitions. Despite their critical importance, sustainability transition research (STR) still lacks comprehensive conceptual work addressing conflicts in sustainability transitions. To bridge this gap, this paper discusses insights from political ecology and Gramscian thought and develops Gramscian political ecology as an integrated framework for examining transition conflicts. Using conflicts as analytical entry points, this framework illuminates the contested nature of sustainability transitions, thereby deepening our understanding of the inherent tensions, contradictions, and obstacles.

This paper begins with a succinct review of STR, acknowledging its advancements in incorporating early critiques (Köhler et al.,

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[1] There is an ongoing debate on the use of the terms 'transition' and 'transformation' in socio-ecological research (Fisher et al., 2022; Hölscher et al., 2018). As this paper contributes to the sustainability transition literature and considering that terminological differences are largely due to different research communities (Hölscher et al., 2018, p. 1), the term 'transition conflict' instead of 'transformation conflict' is used.

https://doi.org/10.1016/j.eist.2024.100812
Received 16 June 2023; Received in revised form 23 December 2023; Accepted 9 January 2024
Available online 18 January 2024
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2019; Truffer et al., 2022) and highlighting its limitations in capturing the contested nature of socio-ecological change. Addressing this gap, the paper introduces a political ecology perspective, putting spotlights on four key research themes: conflicts, nature, power, and justice. Building on these insights, the paper enriches political ecology with a theory of conflict by drawing on the works of Antonio Gramsci and conceptualizing transition conflicts as struggles for hegemony. This synthesis results in an integrated framework of Gramscian political ecology which enables the analysis of struggles for hegemony in sustainability transitions. This includes examining how social forces engage in conflict action to defend fossil fuel hegemonies, promote green capitalism, pursue eco-authoritarianism, or strive for radical socio-ecological transformation. By briefly examining coal transitions in South Africa and Germany, the paper illustrates how Gramscian Political Ecology can be used for empirical analysis. In sum, the paper advances the understanding of transition conflicts, including their configurations, dynamics, and impacts on socio-ecological change, and presents Gramscian political ecology as a non-reductionist, non-essentialist, and context-sensitive framework for empirical analysis.

2. A political ecology perspective on the sustainability transition debate

STR primarily approaches sustainability transitions in terms of socio-technical transitions (Köhler et al., 2019). Most prominently, the Multi-Level Perspective (MLP) posits that transitions in socio-technical systems occur through interactions between niche, regime, and landscape levels (Geels 2002, 2011). In the MLP, low-carbon technologies emerge in societal niches and may be taken up as solutions by incumbent regimes in response to disruptive changes at the landscape level. Early iterations of this model conceptualized sustainability transitions primarily in terms of technological innovations, external shocks, and institutional adaptations.

A vibrant debate between STR and critical social scientists has since emerged (Fisher et al., 2022). An overemphasis on the technological dimension of sustainability transitions has led to criticism that "the socio-' or social element is (…) frequently missing" (Jenkins et al., 2018, p. 67). STR literature has been criticized for conforming to a "decidedly teleological and techno-deterministic perspective" (Lawhon and Murphy, 2012, p. 360) that downplays the messy reality of socio-political contestations and underlying power dynamics. Furthermore, STR's techno-managerial focus is based on Cartesian conceptions of nature as separate from and external to society. Critics argue that this human-nature dualism gives rise to "depoliticized environments" and a "post-political condition" (Swyngedouw, 2011, p. 253), in which environmental problems are mainly addressed through managerial governance and technological innovations, instead of confronting the domination, control, and appropriation of nature in the Anthropocene (Pichler et al., 2017). In addition, postcolonial critics highlight that STR is premised on Western assumptions of modern progress, market institutions, and liberal democracies that do not reflect the realities in the Global South (Wieczorek, 2018). Further criticisms include STR's neglect of colonial histories and their reverberations (Ghosh et al., 2021), post- and decolonial theories (Hopkins et al., 2020), issues of justice (Newell and Mulvaney, 2013), and gender, intersectional, and diverse perspectives (Preuß et al., 2021; Wolfram and Kienesberger, 2023).

Recent contributions to STR have acknowledged and addressed these critiques, thereby sparking lively discussions on power, politics and political economy in sustainability transitions (e.g., Avelino and Wittmayer, 2016; Ciplet and Harrison, 2019; Ford and Newell, 2021; Geels, 2014, 2018; Kivimaa et al., 2021; Krüger and Pellicer-Sifres, 2020; Lawhon and Murphy, 2012; Newell and Mulvaney, 2013; Scheidel et al., 2018; Scoones et al., 2015). While the contested character of sustainability transitions is now widely recognized, there still remains conceptual work to be done. A political ecology perspective could fill this gap by bringing an explicit focus on conflicts, nature, power, and justice to STR. While so far the exchange between political ecology and STR has been limited (e.g., Brand, 2016; Görg et al., 2017; Lawhon and Murphy, 2012; Sovacool, 2021), political ecology offers valuable insights into the transition debate, as succinctly put by the Austrian Panel on Climate Change (2022, p. 3, own translation): "A transformation toward climate-friendly structures is associated with serious conflicts of interests and goals. Understanding and explaining these conflicts and outlining ways of overcoming them is a core task of political ecology." The subsequent sections present four key themes in political ecology: (1) *conflict* as a key mode of sustainability transitions, (2) *nature* as the terrain of conflict, (3) *power* as structuring the conflict terrain, and (4) *justice* as a normative goal and major point of contention. These four themes were chosen because they are key themes in political ecology research, and because they help to address the gaps identified in STR literature and provide insights into the contested nature of sustainability transitions.

2.1. Conflict

Conflicts drive sustainability transitions. Conflict sociologists underscore the key role of social conflicts for social change, arguing that "social change is driven especially by conflict" when "long periods of relatively stable domination are punctuated by intense and dramatic episodes of group mobilization" (Collins, 1990, p. 68). Political ecology highlights the political dimension of environmental issues and thus focuses on environmental conflicts, defined as social conflicts over resources, power, and socio-environmental change (Le Billon 2015). Consequently, it offers a lens through which to approach conflicts accompanying sustainability transitions at different levels of abstraction. At its core, transition conflicts are about trajectories of socio-ecological change. Firstly, transition conflicts emerge when incumbent forces defend fossil fuel-based systems that present "a wildly successful material order, an order that has heretofore been presumed to be net beneficial, salutary, indeed essential and just" (Princen et al., 2015, p. 97). Secondly, transition conflicts arise about different transition pathways between proponents of ecological modernization, radical socio-ecological transformation, and eco-authoritarianism (Adloff and Neckel, 2021). Yet, transition conflicts are not only about broad socio-ecological trajectories but can also emerge from tensions, contradictions, and trade-offs in the implementation of concrete policy measures. Examples include conflicts between social justice and market-oriented transitions (Müller et al., 2020), between energy and climate justice (Newell and Mulvaney, 2013), and between rapid decarbonization on the one hand and concerns of social justice

(Newell et al., 2022), inclusive governance (Ciplet and Harrison, 2019), and coal workers' interests (Kalt, 2021) on the other hand. Political ecologists have identified various conflict types in transition processes, including conflicts between environmental interests and class or labor interests (Dörre et al., 2022; Räthzel and Uzzell, 2012), conflicts arising from differences in culture, values, and lifestyles (Sommer and Schad, 2022), and conflicts over green extractivism in the Global South (Jerez et al., 2021; Kalt et al., 2023). Contrary to conventional views of conflicts as mere obstacles, political ecology sees them not only as barriers to sustainability transitions but also as potential drivers of progressive socio-ecological change. Under favorable conditions, some "conflicts [may] be strong transformative forces for sustainability" (Temper et al., 2018b, p. 575).

In STR, attention to the role of conflicts is increasing. Early foundational work already recognized that "[c]hange arises from conflicts, power struggles, contestations, lobbying, coalition building and bargaining" (Geels and Schot, 2007, p. 415; see also Geels, 2014, 2018). Recent work in STR has shifted focus to the dismantling of carbon-intensive systems through disruption (Kivimaa et al., 2021), discontinuation (Stegmaier et al., 2014), phasing out (Rinscheid et al., 2021), or exnovation (Heyen et al., 2017). Common to this work is an emphasis on conflicts and power because "the users and those who profit from the elements selected for elimination usually have an interest in the perpetuation of the existing patterns of production and consumption" (Krüger and Pellicer-Sifres, 2020, p. 117). Despite STR's growing recognition of the role of conflicts in sustainability transitions, there is a need for further conceptualization of transition conflicts. Political ecology provides a valuable perspective that highlights the role of conflicts in socio-ecological change and offers key insights into nature as the terrain on which transition conflicts unfold.

2.2. Nature

Transition conflicts take place on the terrain of nature and involve the transformation of unsustainable patterns of appropriation, control and exploitation of nature. However, STR lacks ontological reflections on the concepts of nature and society. Nature is frequently equated with 'the environment', thus construing nature as an external space separate from society. From this anthropocentric perspective, nature is perceived as a reservoir of resources and sinks that sets external limits for societal development and that must be aligned with planetary boundaries to guarantee a "safe operating space for humanity" (Rockström et al., 2009, p. 472). Conversely, this view risks homogenizing societies, with their complex relations of power and difference, into an undifferentiated humanity that impacts on nature. This is exemplified by the concept of the Anthropocene, where "humankind has become a global geological force in its own right" (Steffen et al., 2011, p. 843). Within STR, the Cartesian human-nature dualism is particularly prominent in ecomodernist perspectives that advocate for the decoupling of the human economy from the natural environment (Isenhour, 2016).

In contrast, political ecology regards nature and society as inseparably interlinked, a relationship variably understood in terms of hybridity and entanglement (Haraway, 1985), capitalist production of nature (Smith, 2008), or socio-ecological metabolism through which "individuals, groups and cultures shape and regulate their relationships with nature" (Becker et al., 2011, p. 77, own translation; see also Görg et al., 2017). Rather than focusing on techno-managerial interventions to lower atmospheric carbon levels, political ecology emphasizes the reorganization of society's relation with nature. It recognizes that "[w]hat is at stake in the Anthropocene is not so much the technological capacity of humanity to transform the environment but our ability to reshape societal drivers of accelerating resource use, that is, to transform society-nature relations in a more democratic and sustainable way" (Pichler et al., 2017, p. 34). Thus, political ecology argues that the central terrain of transition conflicts is not the natural environment in need of repair but rather society-nature relations that must be transformed through changes in power relations.

2.3. Power

Power dynamics play a crucial role in sustainability transitions, as transformations of society-nature relations occur within the context of uneven power relations, create winners and losers, and often face strong opposition from incumbents. Early critics, however, argued that STR "has fallen short (…) in its understanding of power and political economy" (Scoones et al., 2015, p. 7), limiting the scope of power to state and elite actors (see also Hess, 2014; Feola, 2020; Meadowcroft, 2011). In response, power and agency have emerged as key themes in recent STR (e.g., Avelino, 2017; Avelino and Wittmayer, 2016; Brisbois, 2019; Geels, 2014; Kump, 2023; Raj et al., 2022; Scoones et al., 2015). Although yielding valuable insights, many studies lack an explicit grounding in theories of power, often failing to account for "the basis and (deeper) nature of incumbent power" (Ford and Newell, 2021, p. 2).

Political ecology encompasses diverse perspectives on power, such as actor-oriented, neo-Marxist, and poststructuralist approaches (Svarstad et al., 2018). Actor-oriented political ecology examines how individuals and groups influence socio-ecological change by exerting power, defined as "the ability of an actor to control their own interaction with the environment and the interaction of other actors with the environment" (Bryant and Bailey, 1997, p. 37). Adopting actor-oriented approaches, such as Avelino and Wittmayer's (2016) modification of the MLP for examining power strategies of transition agents at various levels, enables analyses of how actors try to inscribe their interests into sustainability transitions. Moreover, an actor-oriented perspective highlights often-overlooked transition agents, such as grassroots initiatives (Scoones et al., 2015), environmental justice movements (Temper et al., 2018a), labor movements (Kalt, 2022), and feminist movements (Anderson et al., 2019), and their power to shape sustainability transitions.

Neo-Marxist political ecology, on the other hand, "combines the concerns of ecology and a broadly defined political economy" (Blaikie and Brookfield, 1987, p. 17) to examine constraints on agency in the capitalist political economy, with power located in economic structures and class relations. This perspective underscores the role of political-economic structures in shaping the terrain of transition conflicts, by posing structural constraints for sustainability transitions, setting in motion new rounds of green capital accumulation, and perpetuating exploitation and oppression. Echoing the call "for a critical turn to capitalism" in STR (Feola 2020), a

few studies explore how sustainability transitions create new rounds of capital accumulation that lead to new forms of dispossession, inequalities, and conflicts. Examples include lithium extraction in the European periphery (Riofrancos, 2023), climate-smart agriculture in Mozambique (Bruna, 2023), and wind farms in Mexico (Dunlap, 2018).

Poststructuralist perspectives emphasize discursive power as an additional dimension of power. Drawing on Foucauldian notions of discourse, governmentality, and biopolitics, poststructuralist political ecologists argue that the study of power "must include a consideration of the discourses and practices through which nature is historically produced and known" (Escobar, 1996, p. 325). Research might examine how incumbents use "discourses of climate delay" in transition conflicts to impede progress (Lamb et al., 2020), how power is exerted through knowledge-making practices in the governance of sustainability transitions (Jensen et al., 2018), or how the theatrical rituals of climate summits suppress radical critique (Aykut et al., 2021). Taken together, these three perspectives shed light on the intricate power relations in sustainability transitions that both shape the terrain of transition conflicts and are themselves shaped through these conflicts.

2.4. Justice

In a world marked by inequality, sustainability transitions risk reproducing these inequalities. Neglecting social justice in STR tacitly accepts "the risk of 'locking in' patterns of exploitation and dispossession that characterize the current global political economy, even while seeking to overcome carbon 'lock in'" (Healy and Barry, 2017, p. 451). Increasingly, STR is recognizing issues of equity and justice after energy justice scholars have drawn attention to households without access to energy, workers dependent on fossil fuel industries, and communities disproportionately impacted by energy transitions (e.g., Healy and Barry, 2017; Heffron and McCauley, 2018; Jenkins et al., 2018; Newell and Mulvaney, 2013). Recent contributions extend the concept of justice beyond Western liberal paradigms (Álvarez and Coolsaet, 2020; Müller et al., 2022; Tornel, 2023) and integrate multi-species justice perspectives into the study of sustainability transitions (Rastegar, 2022).

The concept of just transition has gained particular prominence in STR, emphasizing the crucial need to address injustices in sustainability transitions: "[I]nattention to justice issues can cause injustices, whereas attention to them can provide a means to examine and potential (*sic*) resolve them" (Jenkins et al., 2018, p. 66). However, the term 'just' is an empty signifier, raising questions about what justice is, who can claim justice, how justice is realized, and what dimensions of justice (i.e. distributive, procedural, recognitional, restorative) should be prioritized. Transition conflicts often revolve around competing justice claims: "Whether the 'just' in just transitions should prioritize the struggles of displaced fossil fuel workers and communities, interstate, interregional or intersectoral inequalities, inter-generational justice or give priority to colonized people against settler colonial states or even the ecological debt between the North and the South, is far from settled" (White, 2020, p. 6). For example, coal workers and unions often define a just transition as a gradual or partial shift away from coal. Consequently, they tend to come into conflict with climate movements that consider such slow transitions as deeply unjust for those most affected by the climate crisis (Kalt, 2021). Overall, political ecology with its normative commitment to "the struggles, interests and plight of marginalized populations" (Bridge et al., 2015, p. 8) is ideally positioned to analyze the ways in which various actors engage in transition conflicts with diverse claims of justice and competing visions of just transition.

To recap, a political ecology perspective broadens the scope of STR, moving beyond socio-technical innovations and institutional adaptations to examine the transformation of society-nature relations in the context of capitalist, colonial and patriarchal power relations. Furthermore, political ecology provides a focus on how these transformations occur through conflicts. The interrelated concepts previously discussed can guide analyses into the contested dynamics of sustainability transitions: (1) Conflicts provide analytical entry points to unpack contested transformations of (2) society-nature relations, taking place in the context of (3) uneven power relations where claims about (4) justice and just transition often play a crucial role. From a political ecology perspective, transition conflicts are struggles on the terrain of nature about how societies respond to ecological crises and re-organize their metabolism with nature to meet social needs under ecological constraints. On an uneven conflict terrain shaped by intersecting power relations, collective actors with conflicting interests, goals, and beliefs struggle over the trajectories of socio-ecological change as well as the distribution of transition costs and benefits, involvement in decision-making processes, and recognition of diverse knowledge, values, and ideas.

3. A Gramscian perspective: transition conflicts as struggles for hegemony

Despite offering valuable insights into the contested dynamics of sustainability transitions, political ecology, as a broad research program rather than a specific theoretical framework, faces limitations in developing a theory of transition conflicts. Conflict has been a core theme in political ecology since the early works of Blaikie and Brookfield (1987) on land conflicts, extending to later studies on conflicts over access to resources (Bryant 1998), the distribution of ecological goods and bads (Martínez-Alier 2002; Scheidel et al., 2018), and cultural representation and recognition (Escobar 2006). However, the concept of conflict at times remains too vague when not explicitly theorized or overly broad when simultaneously encompassing economic contradictions, social divisions, political negotiations, and practices of domination and resistance (Dietz and Engels 2014, p. 81). Gramscian thought can enrich political ecology and STR with a theory of conflict grounded in critical social theory and facilitate the development of a framework and analytical procedure for the study of transition conflicts.

In recent years, a neo-Marxist strand in political ecology has embraced Gramscian thought, asserting that "political ecology should be Marxist, if by Marxist we mean Gramscian" (Mann 2009: 335). A Gramscian-inspired political ecology, discussed in detail below, focuses on how, during ecological crises, nature becomes a terrain for struggle through which hegemony is exercised and contested

(Andreucci et al., 2017; Ekers et al., 2009; Loftus 2015; Mann 2009; Perkins, 2011). Although few, Neo-Gramscian contributions to STR have been notable, exploring issues such as hegemonic power and ideology in sustainability transitions (Scoones et al., 2015), resistance and accommodation regarding fossil fuel hegemonies (Newell, 2019), the contested diffusion of fossil fuel divestment norms (Blondeel, 2019), competing hegemony projects in the European energy transition (Haas, 2019), the reaffirmation or questioning of hegemony in various transition agendas (Hamilton and Ramcilovic-Suominen, 2023), and extending the MLP with a neo-Gramscian framework (Ford and Newell, 2021). Subsequent sections will expand on these lines of inquiry, specifically focusing on how Gramscian thought can contribute to developing a theory of transition conflicts.

By integrating a political ecology perspective into a conceptual framework rooted in critical social theory, this approach facilitates the analysis of the contested dynamics of sustainability transitions. The following sections outline Gramscian political ecology as a framework that merges the four research foci of political ecology – conflicts, nature, power, and justice – with the insights of (neo-) Gramscian hegemony theory. Firstly, transition conflicts are redefined as struggles for hegemony. Secondly, nature is conceptualized as the terrain on which struggles for the hegemony of specific society-nature relations take place. Thirdly, political ecology's three main perspectives on power – materialist, discursive, and actor-centered – are embodied in a Gramscian approach that integrates material structures, discursive hegemony, and agential power. Finally, justice and just transition represent emancipatory ideals reflected in the normative ethos of (neo-)Gramscian theory, which focuses on how hegemonic structures create injustices and how counterhegemonic forces pursue emancipation and justice.

3.1. Struggles for hegemony in times of organic crises

From a Gramscian perspective, transition conflicts can be conceptualized as struggles for hegemony. Much of the thinking of the Italian Marxist Antonio Gramsci revolved around conflicts, as Gramsci believed that "politics occurs in the context of conflict, struggle and contestation" (Fontana, 2006, p. 30). Hegemony, a key concept in his thought, explains how ruling powers maintain control not only through coercion but also by gaining the consent of subordinate groups to relations of domination. This prevents social contradictions, inequalities, and oppressive conditions from erupting into open conflicts. Nevertheless, hegemony is never absolute and must be constantly reproduced. When crises arise, they can deepen the cracks and ruptures in the hegemonic order. Unlike 'conjunctural' crises that can be managed by hegemonic forces, 'organic crises' arise when "incurable structural contradictions have revealed themselves" (Gramsci, 1971, p. 178). Organic crises can destabilize hegemonic formations and create "a new terrain of political and cultural contention and the possibility (but only the possibility) of social transformation" (Carroll, 2010, p. 170–171). In this way, organic crises create the conditions for the emergence of "struggles for hegemony" (Gramsci, 1971, p. 404).

Current interconnected ecological, economic, social and political crises may culminate in organic crises that unsettle the hegemony of fossil capitalism and trigger conflicts about competing transition pathways. While Gramsci and many neo-Gramscian scholars have traditionally neglected ecological issues due to their focus on social structures and class struggle, some political ecologists have begun to explore the ecological dimension of hegemony (Adkin, 2016; Ekers et al., 2009; Loftus, 2015). Gramscian political ecology highlights that transition conflicts take place on the terrain of nature and can therefore be understood in terms of *struggles for the hegemony of specific society-nature relations*. In short, ecological crises transform nature into a key terrain of conflict, which is "a particular terrain over which hegemony is consolidated and contested" (Ekers et al., 2009, p. 289). Social forces seeking hegemony must confront the ecological crisis and "enroll natures and environment (both material and symbolic) in their struggles for hegemony" (Ekers et al., 2009, p. 287). Examining transition conflicts against the backdrop of interconnected crisis tendencies facilitates deeper insights into their causes, dynamics, and transformative potentials.[2]

3.2. Discursive, material, and political dimensions of struggles for hegemony

A Gramscian perspective brings a critical theory of social change to STR that challenges the notion that sustainability transitions are primarily driven by innovations, adaptations, and external shocks. Instead, critical social theory posits that social change emerges from social struggles occurring in the context of structural power relations. These power relations can be understood as "below the surface forces that guide but do not determine historical events" (Marsh and Stoker, 2010, p. 5). From a Gramscian perspective, these 'below the surface forces' can be analyzed in terms of interrelated discursive, material, and political moments, reflecting the main perspective on power in political ecology. Rather than prioritizing one moment over another, Gramscian-inspired research "understand[s] these moments as irreducibly intertwined" (Ekers et al., 2009, p. 288). Examining transition conflicts at the conjuncture of these three interrelated moments provides insights into how historical contexts and power structures shape conflict terrains.

Firstly, struggles for hegemony are discursive struggles over ideology and common sense. Following Stuart Hall (1986, p. 29), Gramsci's concept of ideology consists of "mental frameworks – the languages, the concepts, categories, imagery of thought and the systems of representation – which different classes and social groups deploy in order to make sense of, define, figure out and render intelligible the way society works." Hegemony is attained when subordinate groups internalize the ideologies propagated by dominant

[2] While hegemony is always based on power relations, not all power relations are hegemonic. In contexts where civil society is constrained or faces rapidly shrinking spaces and state or non-state actors rely more on force and coercion than consent, the applicability of a Gramscian framework may seem limited. However, as an anonymous reviewer astutely pointed out, Gramsci himself developed his theories under the repressive regime of fascist Italy. As Gramscian thought highlights the dual nature of power – force and consent – it can also be helpful in analyzing hegemonic rule under oppressive conditions.

social forces, which then naturalize and obscure relations of domination and oppression. In an organic crisis, however, the inherent contradictions within hegemonic ideologies become apparent, raising questions about their naturalness and legitimacy, and triggering struggles over ideologies. In these struggles, social forces engage in a politics of articulation by weaving together disparate discursive elements to construct competing ideologies (Laclau and Mouffe, 1985). These ideological struggles unfold within the sphere of civil society. Contrary to conventional understandings of civil society as an independent third realm alongside the market and state, Gramsci regards civil society as part of the 'integral state' (Gramsci, 1971, p. 12). This is because civil society is the realm in which hegemonic consent is consolidated, thereby complementing the coercive capacities of state power. At the same time, civil society is also a partly autonomous sphere in which hegemonic ideologies can be contested (Perkins, 2011).

Successful ideological struggles result in the sedimentation of ideologies within the fabric of 'common sense'. Common sense refers to an amalgamation of ideas, beliefs, values, norms, traditions, myths, and folk tales that are "inherited from the past and uncritically absorbed" (Gramsci, 1971, p. 333). Due to the multifaceted and inconsistent character of common sense, different ideologies selectively adopt certain elements while discarding others. By anchoring ideologies in common sense, they become more readily accepted as they align with everyday experiences and an immediate sense of reality. This can manifest in subtle yet pervasive ways as "capitalist hegemony is exercised in our lived experiences in virtually unintelligible and therefore surprising ways" (Perkins, 2011, p. 564). Taking this into consideration, a Gramscian analysis of transition conflicts could explore how collective actors engage, both overtly and subtly, in ideological struggles within civil society to gain support for their transition pathways. This analysis would focus on discursive strategies, ideological frameworks, specific narratives, and rhetorical devices, as well as their resonance with common sense.

Secondly, ideologies and common sense are intertwined with the material conditions that shape people's daily lives. Thus, struggles for hegemony also encompass a material dimension. Gramsci (1971, p. 161) asserts that hegemony "must also be economic, must necessarily be based on the decisive function exercised by the leading group in the nucleus of economic activity." However, he diverges from orthodox Marxism by acknowledging that politics and culture cannot be simply reduced to an economic base and instead he emphasizes the dynamic interplay between material, cultural, and political processes.[3] Therefore, ideological struggles take place in the context of "[m]aterial relations [that] provide opportunities and limits within which ideas can develop, without determining them, however" (Pellizzoni and Ylönen, 2016, p. 47). Material relations encompass various aspects, such as the organization of economic life, the development of technologies, and the appropriation of nature. A Gramscian analysis of transition conflicts would need to examine the material conditions that shape conflict terrains and enable and constrain transition pathways. This includes the global political-economic context. For example, Peter Newell's (2021) work on energy transitions focuses on the global history and global political economy of energy production and trade, global energy finance, and global energy governance. Furthermore, the analysis could zoom in on structures and dynamics within regional and national capitalist economies, such as specific accumulation regimes, patterns of production and consumption, labor relations, property relations, infrastructure and technology development, and the regulation of society-nature relations.

Thirdly, hegemony involves not only material and discursive aspects but is also deeply political. Gramsci contends that political agency is crucial for establishing, perpetuating, and dismantling hegemonic orders. Dominant social forces establish and maintain hegemony by coalescing in a 'historical bloc' (Gramsci, 1971, p. 168) and presenting particular interests as universal, forging alliances with powerful groups and individuals, incorporating subordinate groups through concessions and compromise, securing popular consent, and exerting force in 'political society' which encompasses state apparatuses, judicial systems, police, and military. In organic crises, the historical bloc may face challenges to its hegemony when "subaltern segments of society [become] aware of the contradictions within common sense that guide the social relations in which they live" (Perkins, 2011, p. 564). Neo-Gramscian scholars term challenges to hegemony 'counterhegemony', defined as "both the opposition to established apparatuses and practices of domination and the advocacy and construction of emancipatory alternatives" (Carroll, 2016, p. 40).

To examine struggles for hegemony between competing hegemonic and counterhegemonic alliances, neo-Gramscian scholars have developed the concept of 'hegemony project' or 'hegemonic project'[4] (Buckel et al., 2017; Jessop, 1997; Kannankulam and Georgi, 2014). Hegemony projects include a strategy for capital accumulation and involve a set of actors, ideological beliefs, and political strategies. Actors within hegemony projects seek to both implement concrete political initiatives and reconfigure the broader social order through gaining consent and generalizing and institutionalizing their interests (Kannankulam and Georgi, 2014). A Gramscian analysis of transition conflicts could examine how various hegemony projects engage in struggles over socio-ecological futures.

3.3. Strategies in the struggle for hegemony

Gramscian-inspired analyses could examine strategic practices in transition conflicts at different levels of abstraction. While Gramscian thought provides insights into broad strategic modes of (counter-)hegemonic struggles, neo-Gramscian scholars have developed more fine-grained tools for analyzing conflict strategies. Gramsci identifies two key strategic modes in struggles for hegemony: 'war of position' and 'passive revolution' (Gramsci, 1971, pp. 118–120, 238–239). Contrasting with the revolutionary

[3] Laclau and Mouffe (1985) criticize Gramsci for retaining a misguided economic essentialism, while neo-Gramscian scholars counter that Laclau and Mouffe's Gramscian-inspired discourse theory reduces reality to discourse and privileges agency over structure (Carroll and Ratner, 1994; Rustin, 1988). Nevertheless, to avoid the lingering essentialism in Gramscian thought, the focus on the economy as the primary object of struggle and the proletariat as the key historical agent should be reevaluated.

[4] The term 'hegemony project' signifies the pursuit of hegemony, rather than assuming that it has already attained hegemony, as implied by the term 'hegemonic project' used by Bob Jessop (1997), for instance.

strategy of a direct attack on the state through a 'war of manoeuvre' (Gramsci, 1971, pp. 238–239), usually less successful in liberal democracies, war of position aims for radical transformation through prolonged ideological struggles in civil society: "The massive structures of modern democracies (…) constitute for the art of politics as it were the 'trenches' and the permanent fortifications of the front in the war of position: they render merely 'partial' the element of movement which before used to be 'the whole' of war" (Gramsci, 1971, p. 243). In a war of position, the focus is on challenging hegemonic ideologies and spreading counterhegemonic ideas. This entails cultivating ideas about emancipatory change, prefiguring alternative modes of living, and fostering collective identities. Moreover, asserting political and ethical leadership in society, building broad alliances, and linking diverse struggles are key to developing a counterhegemonic 'collective will' (Gramsci, 1971, p. 130) that transcends isolated 'militant particularisms' (Nilsen et al., 2013, p. 81). Moving beyond Gramsci's lingering class reductionism that privileges the working class, a war of position involves forging broad counterhegemonic alliances across intersecting relations of class, race, and gender that challenge "interlocking systems of oppression" (Combahee River Collective, 1977; Simien, 2007).

Contrasting with the war of position as a counterhegemonic strategy from below, passive revolution is a movement from above. It is a response to counterhegemonic challenges through a "process of reformist change from above" (Levy and Egan, 2003, p. 807) that aims at 'trasformismo' (Gramsci 1971, p. 58), referring to "the ability to accommodate pressures for more radical and disruptive change" (Newell 2019, p. 28). This involves integrating subordinate groups and their leaders into the historical bloc, conceding to some demands and "assimilating and domesticating potentially dangerous ideas by adjusting them to the policies of the dominant coalition" (Cox 1983, pp. 166–7). When successful, passive revolution reconfigures social relations to preserve fundamental relations of power and domination and weaken counterhegemonic challenges. This is encapsulated in the dictum by the Italian writer Giuseppe Tomasi di Lampedusa (1958), "everything must change, so that everything can stay the same." Gramscian analyses reveal strategies of passive revolution for resisting and accommodating sustainability challenges and co-opting counterhegemonic challengers (Ford and Newell, 2021; Levy et al., 2016). They also illuminate how counterhegemonic forces engage in war of position, for instance, by forging alliances between progressive governments and social movements within the global climate justice movement (Smith, 2014), building just transition alliances between trade unions and climate activists (Kalt, 2023), or prefiguring counterhegemonic alternatives by integrating commons-based practices into everyday routines (García López et al., 2017).

Beyond the broad strategic modes of war of position and passive revolution, a more fine-grained analysis of conflict strategies requires additional analytical tools. Drawing from neo-Gramscian research in critical migration studies (Buckel et al., 2017, pp. 18–19), the power resource approach in labor studies (Schmalz et al., 2018), and resource mobilization literature in social movement studies (McCarthy and Zald, 1977), various types of power resources can be identified in struggles for hegemony. These include *organizational* resources, encompassing material, financial, and human resources available to organizations, *systemic* resources which involve the ability to cause systemic disruptions, such as through the threat of capital relocation, worker strikes, or activist direct actions, *discursive* resources, referring to the ability to link particular interests to broader discourses and gain consent in public debates, *institutional* resources, including access to the state apparatus and other key institutions, and *coalitional* resources, referring to the ability to pool and mobilize resources from coalition partners. A Gramscian-inspired analysis could examine the conflict strategies that mobilize, deploy, and leverage different power resources in transition conflicts.

4. A Gramscian political ecology of struggles over the hegemony of coal

A brief look at coal transitions in Germany and South Africa serves as an illustrative example of how to apply a Gramscian political ecology framework to explore transition conflicts.[5] The analytical procedure involves three steps: first, examining the historical context and power structures in which transition conflicts are embedded; second, identifying crisis moments that destabilize the hegemonic order and trigger transition conflicts; and third, analyzing collective actors, their interests, and strategies in transition conflicts.

Coal was a major driver of industrial capitalism. During the early industrial revolution, coal-fired steam power provided a reliable and distributable energy source for industries to thrive and intensify their control over both nature and labor (Malm, 2016). Nineteenth-century Germany saw large urban-industrial complexes arise near coal mines, which soon became the birthplace of trade unions and labor parties and a hotbed for socialist and progressive politics, with partially successful labor struggles culminating in a 'carbon democracy' (Mitchell, 2011). Since the late 1950s, coal's dominance has steadily declined because of liberalization and increasing competition with oil and gas and cheaper coal from overseas (Oei et al., 2022, p. 968). Today, coal comprises less than one-third of Germany's electricity mix (German Environment Agency, 2023) and only about 20,000 jobs remain in the coal sector, though with working conditions reminiscent of the Fordist era (Halle Institute for Economic Research, 2018, p. 11). The hegemony of coal in Germany is deeply rooted in the past, that is, in a common sense legacy due to coal's historical role in Germany's rise as an industrial power, post-war economic reconstruction, and European unification (Kalt, 2021). It is also anchored in ideologies of economic prosperity, energy security, and social partnership.

In South Africa, a historical bloc emerged in the early twentieth century with the Minerals-Energy Complex (MEC). The MEC centered around the state-owned electricity monopolist Escom (now: Eskom), which provided cheap electricity from coal mined by 'cheap' Black workers to white-owned mining and industrial capital (Fine and Rustomjee, 1996; Baker et al., 2014; Müller and Claar, 2021). The MEC became further entrenched when international sanctions blocked the Apartheid regime from importing oil, which led

[5] The empirical findings are based on research conducted between 2019 and 2022 in South Africa and Germany by employing a combination of expert interviews, participant observation and document analysis.

to the innovation of coal-to-liquid technologies to circumvent the blockade. In contrast to Germany, South Africa still heavily relies on coal, accounting for 88 % of electricity production and providing around 120,000 jobs (IEA, 2023; TIPS, 2020). During Apartheid, a racialized hegemony of coal emerged through the combination of elements of consent and coercion. The white population, benefiting from access to cheap electricity and believing in ideologies of white superiority and modern progress, consented to this system. Meanwhile, the Black population was coerced into submission through the exploitation of Black mine workers and the designation of Black rural areas as sacrifice zones for coal mining (Finkeldey 2023, p. 35). After Apartheid, the African National Congress (ANC) continued to uphold the hegemony of coal. They now gained the consent of the Black majority population by keeping electricity tariffs low and expanding grid access from less than 50 % to over 80 % by the mid-2000s (Finkeldey 2023, p. 40; World Bank n.d.). These examples illustrate some of the key material and discursive moments that maintained the hegemony of coal in both South Africa and Germany.

The hegemony of coal has been unsettled in the climate crisis, which poses an organic crisis as justifying the reliance on carbon-intensive coal becomes increasingly difficult amidst rapid global heating. In South Africa, the situation is exacerbated by the politicization of a long-looming environmental crisis. Environmental justice groups increasingly struggle against air pollution in the coal mining region of Mpumalanga, one of the world's most polluted areas. In addition, the severe energy crisis that began in 2008 has further destabilized the hegemony of coal with frequent power shortages, rolling blackouts, and rapidly rising energy tariffs. The convergence of these crisis moments has unsettled the hegemony of coal, causing an 'existential crisis' for coal producers and a 'structural crisis' for industries dependent on coal (Malm and Zetkin Collective, 2021, p. 17). These crises reveal contradictions within the hegemony of coal and create conditions for the emergence of transition conflicts over the phase-out of coal and alternative energy futures.

In the context of intersecting crises, three hegemony projects – fossilist, green-liberal, and socio-ecological – compete against each other, albeit with different configurations in South Africa and Germany. Fossilist projects are led by capital factions of coal and energy producers, with large private energy corporations in Germany, and conversely Eskom and private coal mining companies in South Africa. Closely aligned are coal-dependent capital factions, such as the steel and chemical industries in Germany, and the mining and petrochemical industries in South Africa. Fossilist projects aim to preserve the hegemony of coal to protect existing assets and, where possible, to even extend coal use. The strategic mode focuses on delaying the coal transition and seizing opportunities for a fossilist rollback. For instance, Germany's energy crisis, following the Russian invasion of Ukraine, provided an opening for the fossilist project to reactivate mothballed coal-fired power stations. Likewise, South Africa's coal industry seeks to exploit the domestic energy crisis to prevent the decommissioning of coal-fired power stations. Key strategies involve leveraging coalitional power, as coal and coal-dependent capital factions form cross-class alliances with workers and trade unions in the coal sector in both Germany and South Africa. Integrating unions into hegemonic alliances enables fossilist factions to generalize their interests in broader society while simultaneously fostering divisions among progressive forces along a jobs vs. environment divide (Kalt, 2021, 2023; Räthzel et al., 2018). Fossilist factions also rely on institutional and organizational power to exert influence in state apparatuses and gain support from political elites, as seen in their close ties with South Africa's Department of Mineral Resources and Energy and Germany's Social Democratic Party. In addition, fossilist actors leverage their discursive power in ideological struggles by promoting "discourses of climate delay" (Lamb et al., 2020). In Germany, fossilist narratives highlight challenges to energy security, energy affordability, and economic competitiveness and position coal as a 'bridging technology' for the energy transition. In South Africa, fossilist narratives foreground energy security and job losses, while promoting 'clean coal' technology and framing the energy transition as imperialist propaganda.

Fossilist projects face opposition from green-liberal hegemony projects that encompass sustainability-oriented capital factions in the energy, technology, and finance sectors. Pursuing a Gramscian passive revolution, green-liberal projects seek to renew hegemonic power structures by accommodating demands for climate action and energy security while simultaneously creating new opportunities for 'green' capital accumulation through market-driven and finance-led energy transitions. Germany's energy transition, initially rooted in principles of decentralization and community ownership, has been transformed in the past decade towards market integration and centralization by a green-liberal hegemony project (Haas 2017: 166–173). Strategies focus on leveraging institutional power via close ties with the Green Party, presently in the German Government, and on building coalitional power through gaining the support of NGOs and parts of the climate movement. Strategies also involve building discursive power by popularizing an ideology of green growth that modifies and reaffirms the dominant growth ideology, specifically by promoting green technology exports as a means to strengthen Germany's export-oriented accumulation regime. This discursive strategy aligns with deeply ingrained beliefs in common sense regarding economic progress and technological salvation through economic growth (Schmelzer, 2016). In South Africa, a green-liberal hegemony project emerged amid the energy crisis and has achieved some success in promoting a market-driven energy transition by advocating for the liberalization of the energy sector and influencing government policies to adopt a market-based competitive bidding mechanism for renewable energy projects. Discursive strategies encompass promoting a free-market ideology that depicts the market-driven expansion of renewable energy as an efficient, inexpensive, and sustainable solution to an inefficient, corrupt, and polluting state-owned energy sector, which fails to deliver stable and affordable energy (Bode 2013, 104–105). Additionally, strategies draw on development discourses, with some concessions for socio-economic development and local ownership requirements within the government's renewable energy program (Müller and Claar, 2021).

Radical socio-ecological projects oppose both fossilist and green-liberal projects. Aiming to achieve rapid decarbonization, democratization, and socialization of the energy sector, these projects pursue counterhegemonic transformations beyond 'green' capitalism and 'green' colonialism. By employing the strategic mode of a Gramscian war of position, social movements challenge hegemonic ideologies, prefigure alternative energy futures, and build collective identities, movements, and alliances. A key strategy involves building discursive power. In South Africa, narratives focus on mobilizing justice claims linked to demands for the social ownership of renewable energy, reparations for climate debt, and a just transition for coal workers and affected communities. In

Germany, climate justice movements attempt to delegitimize coal power and challenge hegemonic growth ideologies propagated by fossilist and green-liberal projects, instead promoting alternative degrowth ideologies rooted in common-sense values of community, solidarity and care. Furthermore, social movements try to leverage systemic power through disruptive actions, exemplified by those carried out by activist groups Ende Gelände, Extinction Rebellion, and Letzte Generation in Germany. In addition, they build coalitional power, as seen in alliances between environmental justice organizations, community groups, and trade unions in the Life After Coal campaign and the Climate Justice Coalition in South Africa. In sum, these brief examples illustrate how Gramscian political ecology serves to contextualize transition conflicts historically, explain their emergence in moments of crisis, and explore how these conflicts develop through strategic interactions among competing hegemony projects.

5. Conclusion: A Gramscian political ecology of transition conflicts

Conflicts play a crucial role in sustainability transitions and may act as catalysts or barriers. This paper addressed gaps in STR relating to the contested nature of sustainability transitions by drawing on political ecology's insights on conflict, nature, power, and justice. In redefining these concepts through the lens of Gramscian hegemony theory, this paper developed a Gramscian political ecology that combines the strengths of both political ecology and Gramscian thought. Within the framework of Gramscian political ecology, transition conflicts were conceptualized as struggles for hegemony on the terrain of society-nature relations in the context of ecological crises. The brief case studies of coal transitions in South Africa and Germany illustrated the analytical value of Gramscian political ecology for the empirical study of conflicts in sustainability transitions.

Gramscian political ecology offers an analytical framework for examining how various actors, both dominant and marginalized, struggle for socio-ecological futures that either preserve or challenge hegemonic orders. This framework yields key insights for analyzing sustainability transitions. First, it uses conflicts as analytical entry points to unravel the contested dynamics of sustainability transitions. It specifically focuses on how collective actors enroll nature in their narratives, seek consent for their socio-ecological visions, ideas, and practices, and attempt to anchor these in political institutions and common sense. Second, Gramscian political ecology underscores the importance of contextualizing and historicizing transition conflicts. Doing so helps to comprehend conflict dynamics against the backdrop of structural power relations and historical conditions, which either constrain or enable certain transition pathways. Third, Gramscian political ecology brings into view the agency of marginalized groups and often overlooked transition agents, like climate justice movements, trade unions, and grassroots initiatives. It acknowledges their pivotal roles in sparking transition conflicts, contesting power relations, and steering sustainability transitions towards just outcomes. Lastly, its normative commitment to justice, equality, and emancipation facilitates a critique of hegemonic power relations, ideologies, and injustices. It highlights the radical transformative potentials of counterhegemonic ideas and practices emerging in transition conflicts and explores potentials, barriers, and challenges for realizing just transitions.

As transition conflicts are likely to multiply and intensify as the ecological crisis escalates, a deeper understanding of their dynamics is crucial. In response, this paper presented a framework for analyzing transition conflicts, thereby contributing to the theoretical debate on conflicts, power, and justice in sustainability transitions as well as laying the groundwork for empirical research on conflicts in sustainability transitions. Future research could include a deeper focus on underrepresented transition agents, building on existing research on the roles of social movements and trade unions in transition conflicts (Finkeldey 2023; Piggot 2018; Stevis and Felli 2015; Temper et al., 2018a), or the strategies of the far right which increasingly promote eco-authoritarian responses to the climate crisis (Forchtner 2020; Malm and Zetkin Collective 2021). In addition, comparative research employing a Gramscian political ecology lens could systematically assess the impact of transition conflicts on transition outcomes and expose the context-specific conditions under which transition conflicts become either catalysts or barriers in sustainability transitions. Lastly, integrating post- and decolonial theories with Gramscian political ecology could make the framework better attuned to studying transition conflicts in the Global South and across the global North-South divide.

CRediT authorship contribution statement

Tobias Kalt: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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